

PIC18F27/47Q10 Silicon Errata and Data Sheet Clarifications

PIC18F27/47Q10



The PIC18F27/47Q10 devices you have received conform functionally to the current device data sheet (DS40002043F), except for the anomalies described in this document.

The silicon issues discussed in the following pages are for silicon revisions with the Device and Revision IDs listed in the table below.

The errata described in this document will be addressed in future revisions of the PIC18F27/47Q10 silicon.

Note: This document summarizes all silicon errata issues from all silicon revisions, previous and current.

Table 1. Silicon Device Identification

Part Number	Device ID	Revision ID	
		A4	B2
PIC18F27Q10	0x7100	0xA004	0xA042
PIC18F47Q10	0x70E0	0xA004	0xA042



Important: Refer to the **Device/Revision ID** section in the current “**PIC18F2X/4XQ10 Memory Programming Specification**” (DS40001874) for more detailed information on Device Identification and Revision IDs for your specific device.

Table 2. Silicon Issue Summary

Module	Feature	Item No.	Issue Summary	Affected Revisions	
				A4	B2
Electrical Specifications	Temperature range	1.1.1	Industrial temperature range only	X	
	Sleep current	1.1.2	Higher current after DFM write operation	X	
Resets	RMCLR flag	1.2.1	POR may clear the RMCLR bit by mistake	X	X
	LPBOR	1.2.2	Trip point rises with temperature	X	
Complimentary Waveform Generator (CWG)	Auto-shutdown sources	1.3.1	CLC2 and CLC6 not available	X	
Analog-to-Digital Converter with Computation (ADCC)	FVR reference	1.4.1	Missing codes when using FVR as reference	X	X
	Burst average	1.4.2	ADCNT may not increment	X	
	ADCRC (FRC) oscillator	1.4.3	Oscillator continues to run in Sleep mode after conversion	X	
	Input slew rate	1.4.4	Unreliable conversion results with fast falling slew rate	X	X
Windowed Watchdog Timer (WWDT)	Window operation	1.5.1	The window feature of WWDT does not operate correctly in Doze mode	X	
Non-Volatile Memory (NVM)	NVMERR	1.6.1	The NVMERR bit is set by device Reset after being cleared by the software	X	
	Self-writes	1.6.2	Do not write above 85°C		X
Oscillator	HFINTOSC	1.7.1	5% variation over temperature range	X	
	XT mode	1.7.2	Maximum clock frequency limited to 2 MHz for XT mode	X	
In-Circuit Debug	Software breakpoints	1.8.1	Software breakpoints are not available	X	X
Program Flash Memory (PFM)	Self-Write	1.9.1	The first instruction following a self-write instruction may not execute	X	X
Note: Only issues indicated in the last column apply to the current silicon revision.					

1. Silicon Errata Issues

NOTICE

This document summarizes all silicon errata issues from all revisions of silicon, previous and current. Only the issues indicated by the bold font in the following tables apply to the current silicon revision.

1.1 Module: Electrical Specifications

1.1.1 Industrial Temperature Range Only

Extended temperature range devices are not released.

Work around

Operate at or below 85°C.

Affected Silicon Revisions

A4	B2
X	

1.1.2 Sleep Current - Higher Sleep Current After Data Flash Memory (DFM) Write Operation

During a DFM write operation while in Sleep mode, the system clock will remain in the Active state. Consequently, once the write operation is completed, the device will exhibit an increased Sleep current if it continues to stay in this state.

Work around

Upon completing the DFM write operation, wake the device up from Sleep mode and re-execute a new Sleep command.

Affected Silicon Revisions

A4	B2
X	

1.2 Module: Resets

1.2.1 The $\overline{\text{RMCLR}}$ Flag in the PCON0 Register Cleared by Mistake

On an initial power-up of the device or when executing a software Reset, the $\overline{\text{RMCLR}}$ flag in the PCON0 register may be improperly cleared by a Power-on Reset (POR) or software Reset ($\overline{\text{RI}}$), thereby indicating a false MCLR event.

Work around

None.

Affected Silicon Revisions

A4	B2
X	X

1.2.2 Low-Power Brown-out Reset (LPBOR) Mode

The Brown-out Reset trip level increases proportionally with temperature to a level where BOR is never released. LPBOR cannot be used reliably because the trip level relative to temperature is indeterminate.

Work around

Use the Normal-Power BOR mode.

Affected Silicon Revisions

A4	B2
X	

1.3 Module: Complementary Waveform Generator (CWG)

1.3.1 CWG Auto-Shutdown Sources

Shutdown sources AS6E (CLC2_out) and AS7E (CLC6_out) are unavailable.

Work around

Route the Configurable Logic Cell (CLC) output through Peripheral Pin Select (PPS) to an output pin, and use the AS0E source selection (the pin selected by CWGxPPS) and PPS controls to select the same pin as the shutdown source.

Affected Silicon Revisions

A4	B2
X	

1.4 Module: Analog-to-Digital Converter with Computation (ADCC)

1.4.1 Missing Codes with FVR Reference

Using the FVR as the positive voltage reference for the ADC can cause an increase in missing codes.

Work around

Method 1:

Increase the bit conversion time, known as T_{AD} , to 8 μ s or higher.

Method 2:

Use V_{DD} as the positive voltage reference to the ADC.

Affected Silicon Revisions

A4	B2
X	X

1.4.2 ADCC Burst Average Mode

When the ADCC is operated in Burst Average mode ($ADMD = 0b011$ in the $ADCON2$ register) while enabling noncontinuous operation and double-sampling ($ADCONT = 0$ in the $ADCON0$ register and $ADDSEN = 1$ in the $ADCON1$ register), the value in the $ADCNT$ register does not increment beyond '0b1' toward the value in the $ADRPT$ register.

Work around

When operating the ADCC in Burst Average mode with double-sampling, enable continuous module operation ($ADCONT = 1$ in the $ADCON0$ register) and set the Stop-on-Interrupt bit (the $ADS0I$ bit in the $ADCON3$ register). After the interrupt occurs, perform appropriate threshold calculations in the software and retrigger ADCC as necessary.

Alternatively, if the CPU is in Low-Power Sleep mode, the ADCC in noncontinuous Burst Average mode can be operated with a single ADC conversion ($ADDSEN = 0$ in the $ADCON1$ register). Doing so compromises noise immunity for lower power consumption by preventing the device from waking up to perform threshold calculations in the software.

Affected Silicon Revisions

A4	B2
X	

1.4.3 ADCRC (FRC) Oscillator Operation in Sleep Mode

If the part is in Sleep mode and the ADCRC (FRC) oscillator is used as clock source to the ADC, the oscillator continues to run after the conversion is complete, increasing the current consumption in Sleep mode. The oscillator will stop after the device exits Sleep mode and resumes normal code execution.

Work around

None.

Affected Silicon Revisions

A4	B2
X	

1.4.4 Unreliable Conversion Results with Fast Falling Slew Rate

When the ADC input falls by greater than 3.2V, with a slew rate faster than -11 V/μs, the following ADC conversion will have the Most Significant bit (MSb) improperly set. This is likely to happen when the ADC input channel is switched from one with a high-level input to another with a low-level input.

Work around

When switching between input channels, discard the first conversion result after the switch. Subsequent conversions will not be affected.

Affected Silicon Revisions

A4	B2
X	X

1.5 Module: Windowed Watchdog Timer (WWDT)

1.5.1 Window Operation in Doze Mode

When enabling the Windowed operation mode in Doze mode, a window violation error is issued even though the window is open and armed. This condition occurs only when the window size is set to a value other than 100% open.

Work around

Method 1:

Use the Windowed operation mode in any mode other than Doze. If disabling the Doze mode is not an option, use the WWDT module without enabling the window.

Method 2:

If the device is in Doze mode, perform the arming process for the window in Normal mode and return to the Doze mode.

Method 3:

If there is an Interrupt Service Routine (ISR) in the application code, the arming within the window can be done inside the ISR with the ROI bit of the CPUDOZE register being set.

Affected Silicon Revisions

A4	B2
X	

1.6 Module: Nonvolatile Memory (NVM)

1.6.1 NVMERR

When a Reset is issued while an NVM high-voltage operation is in progress, the NVMERR bit in the NVMCON0 register is set as expected. After clearing the NVMERR bit, if a Reset reoccurs, the NVMERR bit is set again regardless of whether an NVM operation is in progress or not. A successful write operation will clear the NVMERR condition.

Work around

None.

Affected Silicon Revisions

A4	B2
X	

1.6.2 PFM Writes Above 85° Celsius

Do not perform write operations on the Program Flash Memory (PFM) when the temperature exceeds 85°C.

Work around

Perform PFM writes below 85°C.

Affected Silicon Revisions

A4	B2
	X

1.7 Module: Oscillator (OSC)

1.7.1 Internal HFINTOSC Oscillator Varies up to 5%

The internal HFINTOSC oscillator frequency varies up to 5% over the voltage and temperature range.

Work around

For systems requiring more precision, use an external crystal or ceramic resonator in one of the external oscillator modes.

Affected Silicon Revisions

A4	B2
X	

1.7.2 Maximum Clock Frequency Limited to 2 MHz for XT Mode

The maximum clock frequency for the intermediate gain setting that supports quartz crystal and ceramic resonator operation (XT mode) is being reduced from 4 MHz to 2 MHz.

Work around

For crystal or resonator frequencies above 2 MHz, use HS mode.

Affected Silicon Revisions

A4	B2
X	

1.8 Module: In-Circuit Debug

1.8.1 Software Breakpoints Are Not Available

When debugging code, software breakpoints will not be available.

Work around

None.

Affected Silicon Revisions

A4	B2
X	X

1.9 Module: Program Flash Memory (PFM)

1.9.1 The First Instruction Following a Self-Write Instruction May Not Execute

When performing a self-write operation to Program Flash Memory, the first instruction following a self-write instruction may not execute.

Work around

Add two all-zero `NOP()` macros immediately after the self-write instruction. The all-zero `NOP()` macros are included automatically when the `#include <xc.h>` directive is used in user software.

Affected Silicon Revisions

A4	B2
X	X

2. Data Sheet Clarifications

Note the following typographic corrections and clarifications for the latest version of the device data sheet (DS40002043F):

Note:

Corrections are shown in **bold**. Where possible, the original bold text formatting has been removed for clarity.

2.1 None

There are no known data sheet clarifications as of this publication date.

3. Appendix A: Revision History

Doc Rev.	Date	Comments
G	05/2024	Removed Data Sheet Clarifications and updated Data Sheet Revision. Minor editorial corrections.
F	11/2022	Added silicon erratum item 1.9.1.
E	07/2021	Removed silicon erratum item 1.4.4. Minor editorial corrections.
D	01/2021	Added silicon erratum item 1.8.1. Minor editorial corrections.
C	09/2020	Added new silicon Rev B2 and silicon erratum item 1.6.2
B	08/2020	Updated Table 1, Data Sheet Clarification section. Added silicon erratum items 1.2.2 and 1.7.2. Minor editorial corrections.
A	04/2019	Initial document release

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